LAPORAN TUGAS
PEMOGRAMAN PERANGKAT MOBILE

Disusun oleh:

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C

TEKNIK INFORMATIKA

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TUGAS 1

KALKULATOR BANGUN RUANG

A. Source Code

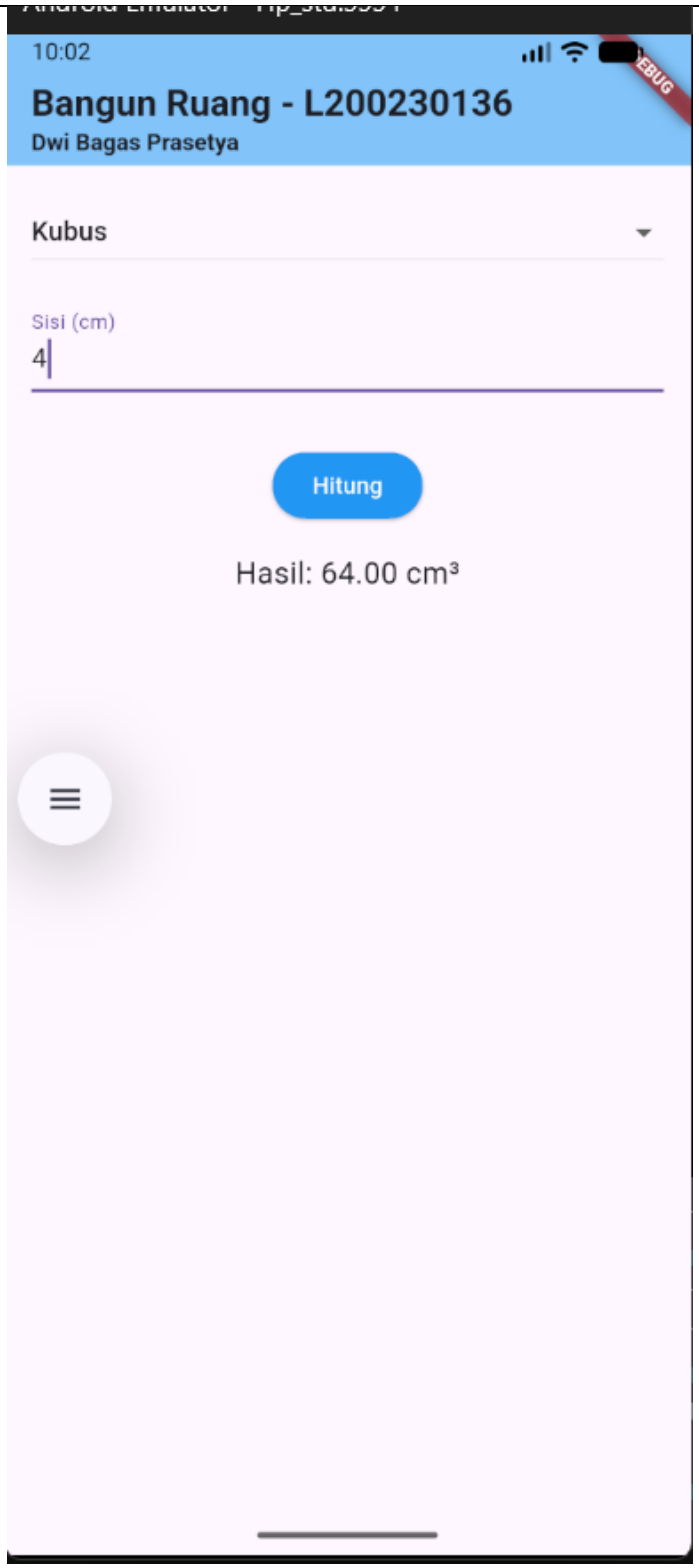
```
import 'package:flutter/material.dart';
import 'dart:math';

void main() {
  runApp(MyApp());
}

// Widget utama aplikasi
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: VolumePage(),
    );
  }
}

// StatefulWidget karena data (hasil) bisa berubah
class VolumePage extends StatefulWidget {
  const VolumePage({super.key});

  @override
  _VolumePageState createState() => _VolumePageState();
}

// Class state yang berisi logika dan tampilan
class _VolumePageState extends State<VolumePage> {
  String selectedBangun = "Kubus";
  double hasil = 0;

  // Controller untuk mengambil input dari TextField
  final TextEditingController controller1 = TextEditingController();
  final TextEditingController controller2 = TextEditingController();

  // key form untuk mem-validasi input
  final GlobalKey<FormState> _formKey = GlobalKey<FormState>();

  @override
  void dispose() {
    // bersihkan controller ketika State dihancurkan
    controller1.dispose();
```

```
    controller2.dispose();
    super.dispose();
  }

  // Fungsi untuk menghitung volume
  void hitungVolume() {
    if (!_formKey.currentState!.validate()) return;

    // Mengambil nilai dari input dan mengubah ke double
    double nilai1 = double.parse(controller1.text);
    double nilai2 =
        double.parse(controller2.text.isEmpty ? '0' :
controller2.text);

    setState(() {
      if (selectedBangun == "Kubus") {
        hasil = pow(nilai1, 3).toDouble(); // Rumus: sisi³
      } else if (selectedBangun == "Tabung") {
        hasil = pi * pow(nilai1, 2) * nilai2; // Rumus: π × r² × t
      } else if (selectedBangun == "Bola") {
        hasil = (4 / 3) * pi * pow(nilai1, 3); // Rumus: 4/3 × π × r³
      }
    });
  }

  @override
  Widget build(BuildContext context) {
    // Scaffold adalah kerangka utama halaman
    return Scaffold(
      appBar: AppBar(
        backgroundColor: const Color.fromARGB(255, 130, 196, 196, 250),
        title: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              "Bangun Ruang - L200230136",
              style: TextStyle(fontWeight: FontWeight.bold),
            ),
            Text(
              "Dwi Bagas Prasetya",
              style: TextStyle(fontSize: 14, fontWeight:
FontWeight.bold),
            ),
          ],
        ),
      ),
      body: Padding(
        padding: EdgeInsets.all(16),
```

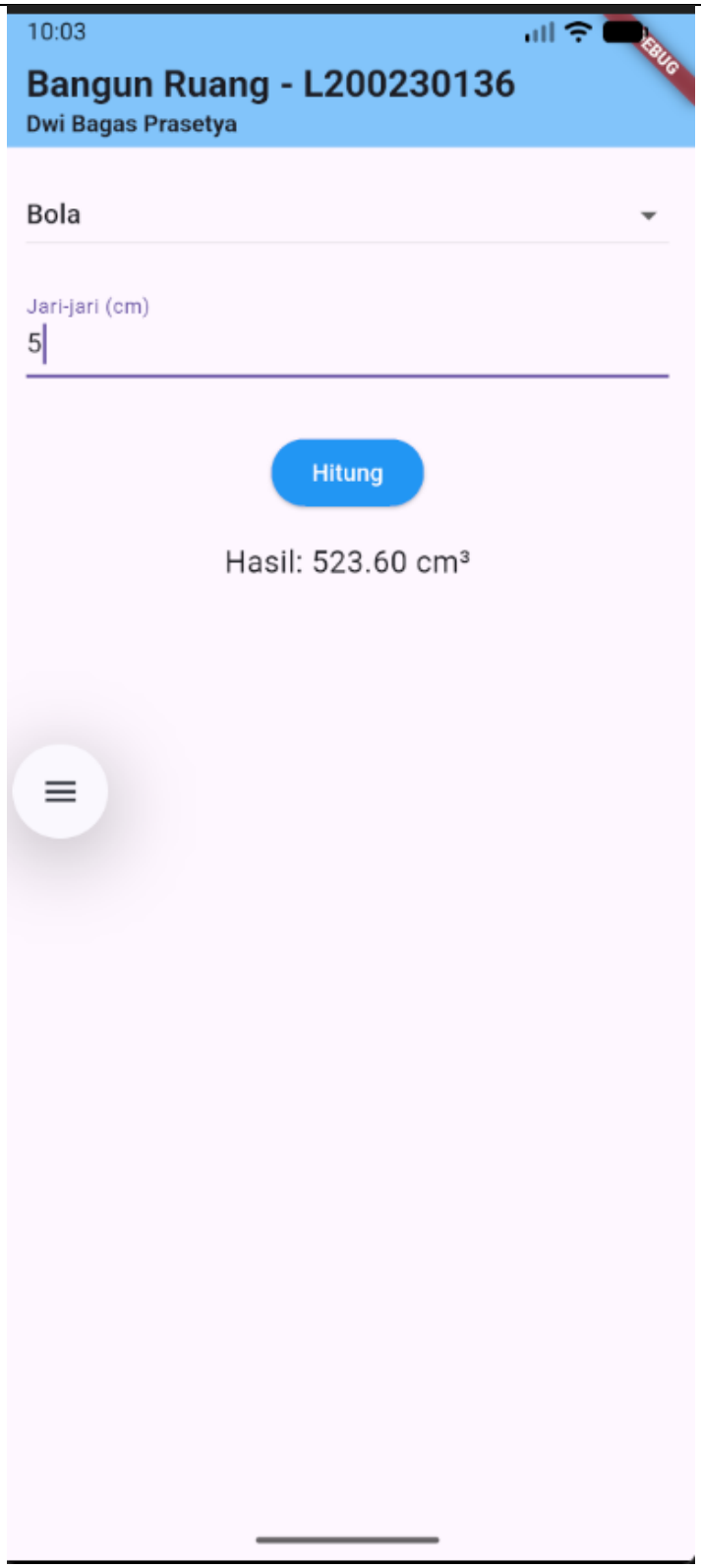
```
            child: Form(
              key: _formKey,
              child: Column(
                children: [
                  // Dropdown untuk memilih bangun ruang
                  DropdownButton<String>(
                    value: selectedBangun,
                    isExpanded: true,
                    items: ["Kubus", "Tabung", "Bola"]
                        .map((bangun) => DropdownMenuItem(
                              value: bangun,
                              child: Text(bangun),
                            ))
                        .toList(),
                    onChanged: (value) {
                      setState(() {
                        selectedBangun = value!;
                        hasil = 0;
                        controller1.clear();
                        controller2.clear();
                      });
                    },
                  ),

                  SizedBox(height: 16), // Jarak antar widget

                  // Input sisi atau jari-jari
                  TextFormField(
                    controller: controller1,
                    keyboardType: TextInputType.number,
                    decoration: InputDecoration(
                      labelText: selectedBangun == "Kubus"
                          ? "Sisi (cm)"
                          : "Jari-jari (cm)",
                    ),
                    validator: (value) {
                      if (value == null || value.isEmpty) {
                        return 'Masukkan nilai';
                      }
                      if (double.tryParse(value) == null ||
                          double.parse(value) <= 0) {
                        return 'Masukkan angka lebih besar dari nol';
                      }
                      return null;
                    },
                  ),

                  SizedBox(height: 16),
```

```
                // Input tinggi hanya muncul jika memilih Tabung
                if (selectedBangun == "Tabung")
                  TextFormField(
                    controller: controller2,
                    keyboardType: TextInputType.number,
                    decoration: InputDecoration(
                      labelText: "Tinggi (cm)",
                    ),
                    validator: (value) {
                      if (value == null || value.isEmpty) {
                        return 'Masukkan tinggi';
                      }
                      if (double.tryParse(value) == null ||
                          double.parse(value) <= 0) {
                        return 'Masukkan angka lebih besar dari nol';
                      }
                      return null;
                    },
                  ),

                SizedBox(height: 16),

                // Tombol untuk menghitung volume
                ElevatedButton(
                  style: ElevatedButton.styleFrom(
                    backgroundColor: Colors.blue, // background
                    foregroundColor: Colors.white, // text
                  ),
                  onPressed: hitungVolume,
                  child: Text("Hitung"),
                ),

                SizedBox(height: 16),

                // Menampilkan hasil perhitungan
                Text(
                  "Hasil: ${hasil.toStringAsFixed(2)} cm³",
                  style: TextStyle(fontSize: 18),
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

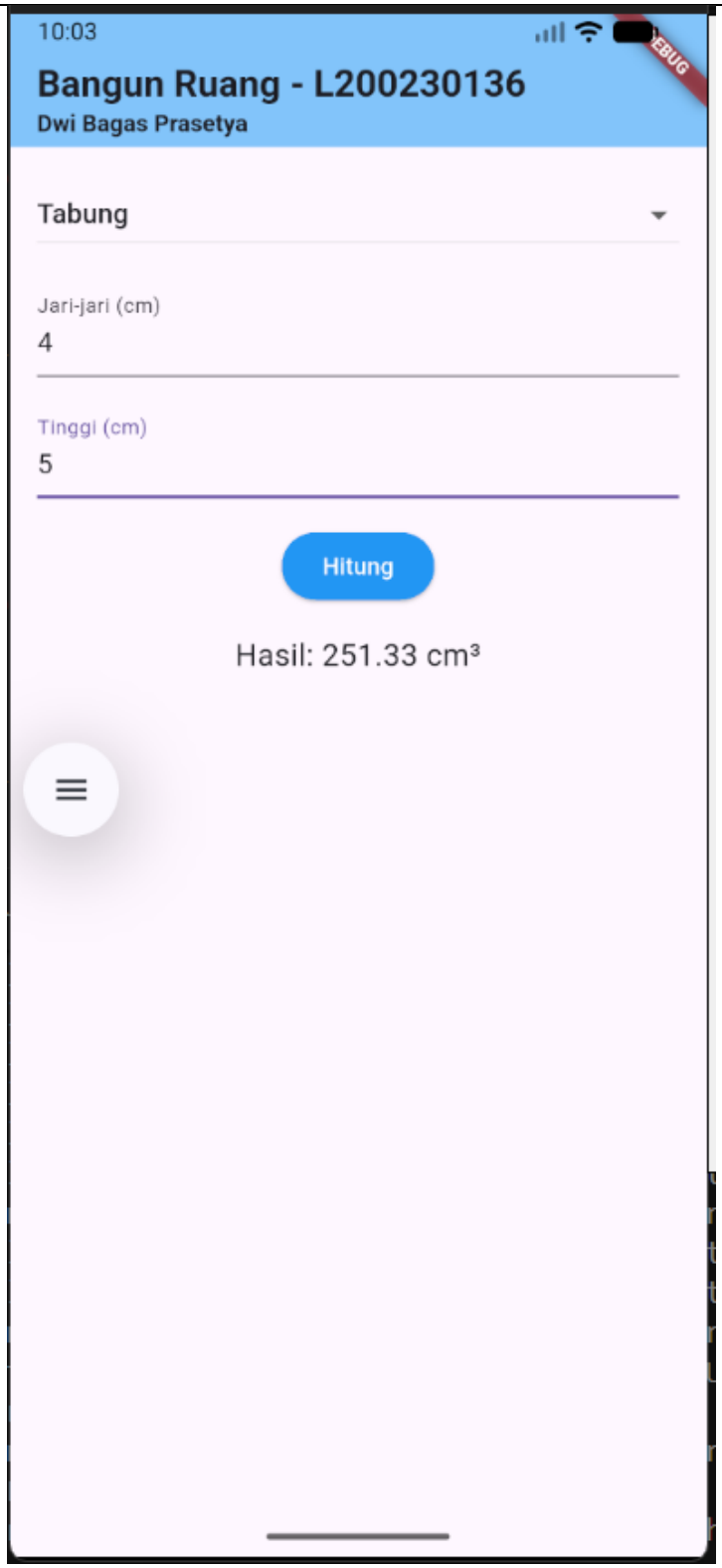
B. Dokumentasi



Gambar 1 Hasil Perhitungan Volume Kubus



Gambar 2 Hasil Perhitungan Volume Bola



Gambar 3 Hasil Perhitungan Volume Tabung

C. Link

[Gazzzee/Tugas1_AppsBangunRuang_PPM_C]